

BMB547

Rates of Change

Anvendelser af matematik i
lifescience

Topics

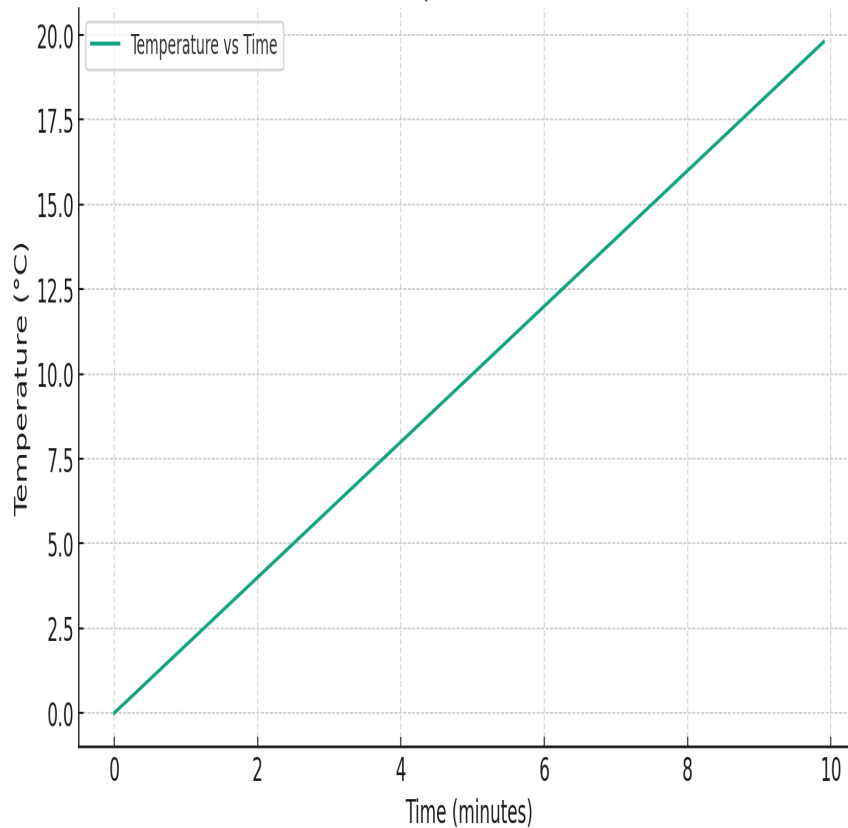
- Gennemsnitlig ændringshastighed
- Enzymkinetik

Verden ændrer sig.

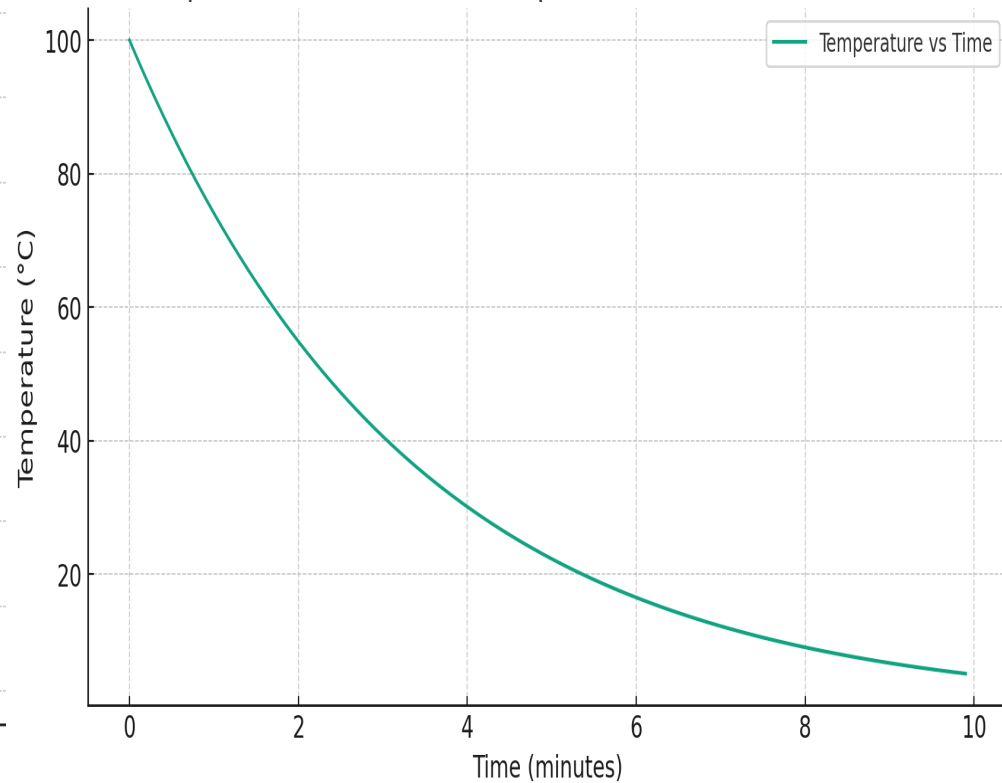
- Ændringshastigheden er en vigtig karakteristik af mange systemer.
- Ting kan ændre sig som en funktion af tid.
- Mange ting ændrer sig som funktioner af for eksempel koncentration, temperatur, tryk...

Opvarmning og køling

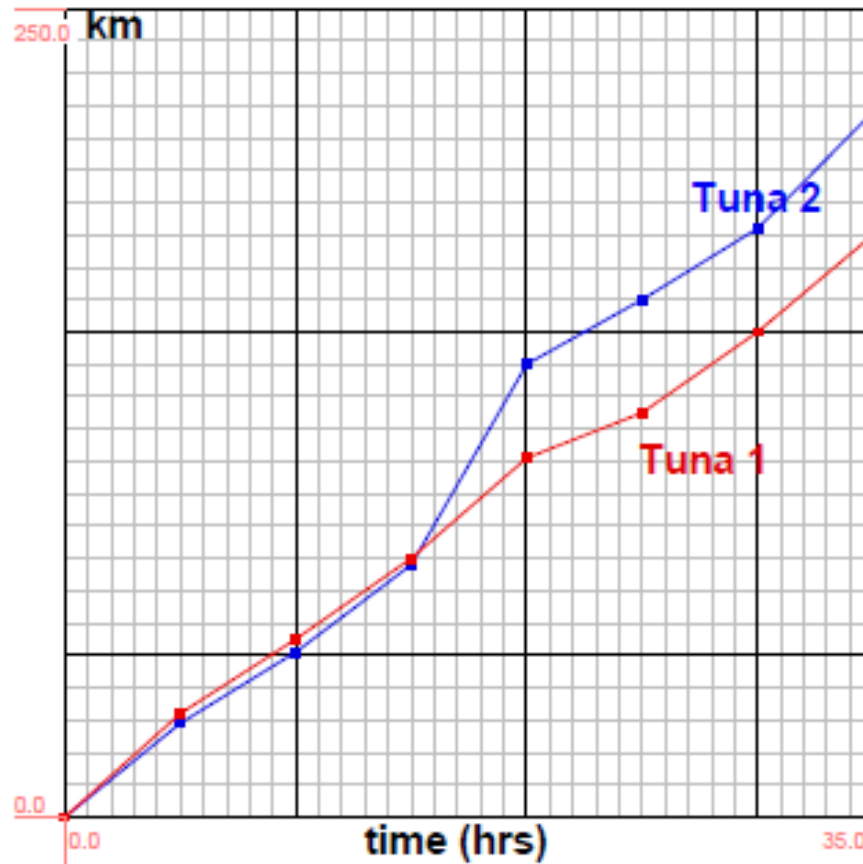
Linear Increase in Temperature as a Function of Time



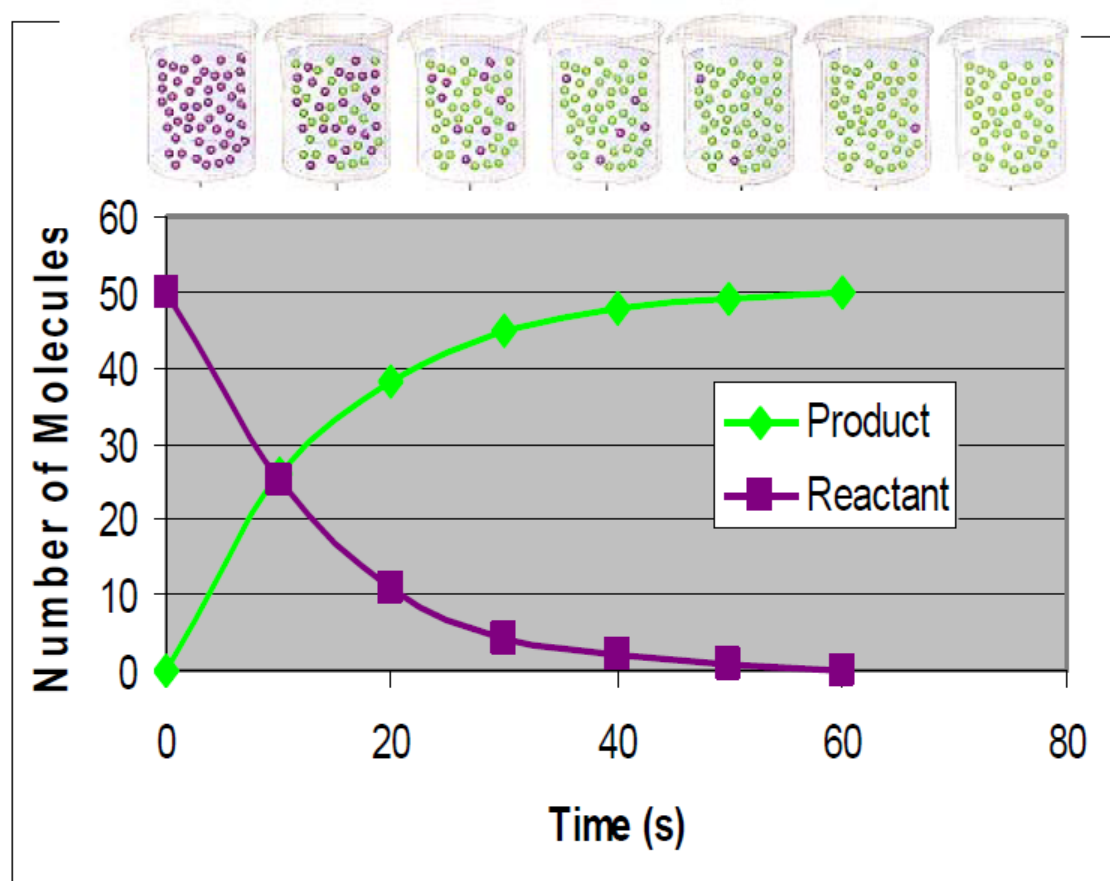
Exponential Decrease in Temperature as a Function of Time



Afstand rejst af to blåfinnet tun over 35 timer

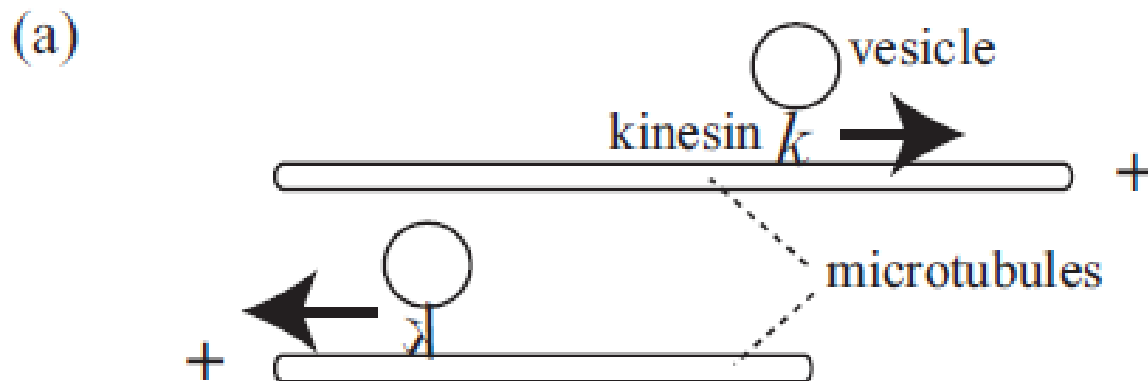


Kemiske reaktioner

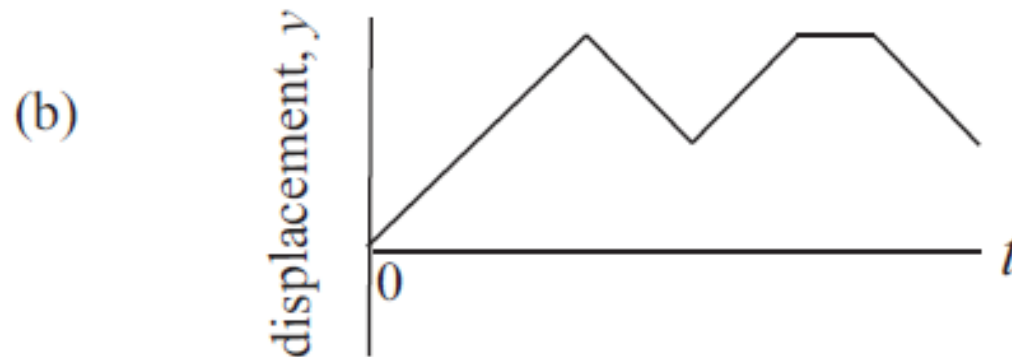
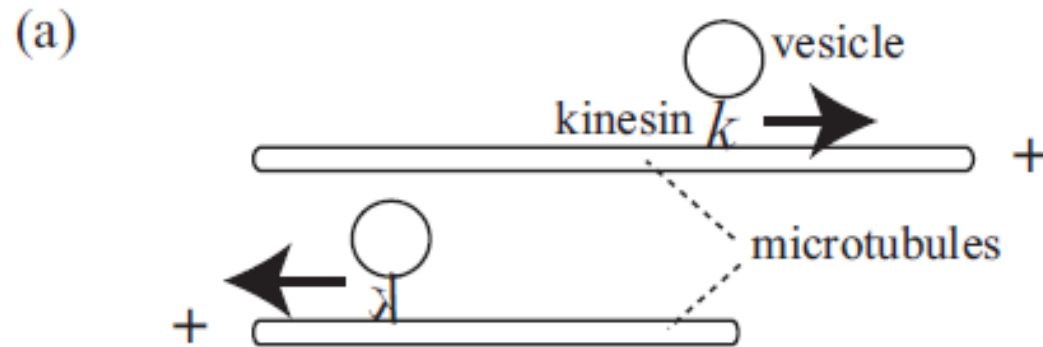


Molekylære motorer

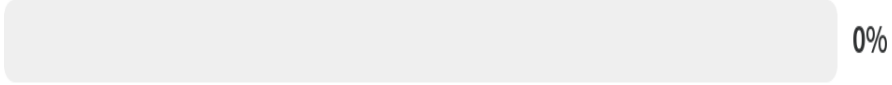
Nogle proteiner fungerer som "molekylære motorer", de transporterer last (vesikler) langs mikrotubuli i en celle. Diagram af kinesin (repræsenteret ved bogstavet k), en motor der bevæger sig mod mikrotubulens ende



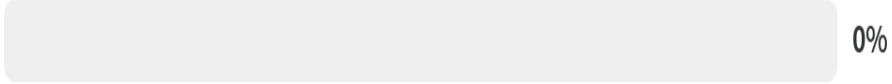
- Den følgende graf viser forskydningen (y) af en vesikel som en funktion af tid.



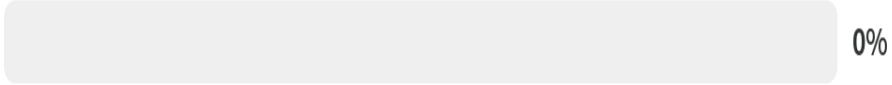
A)



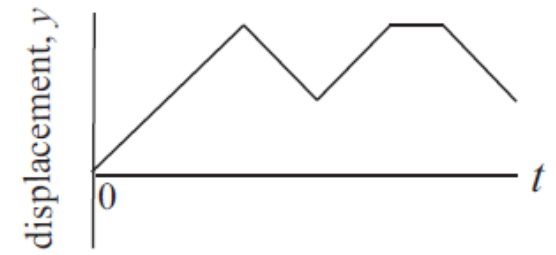
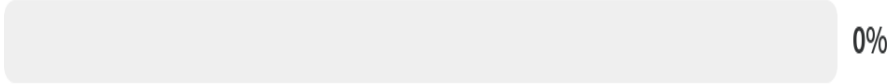
B)



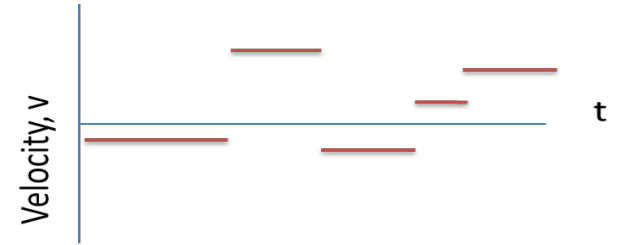
C)



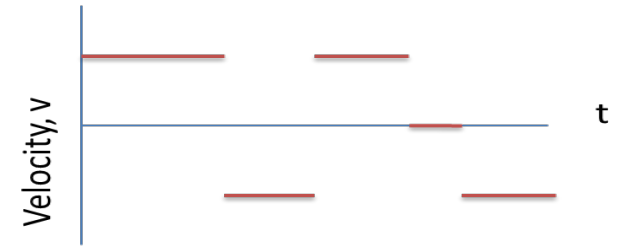
D)



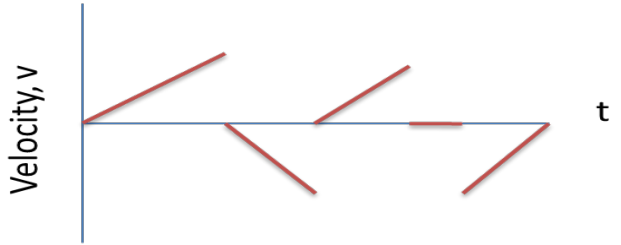
A)



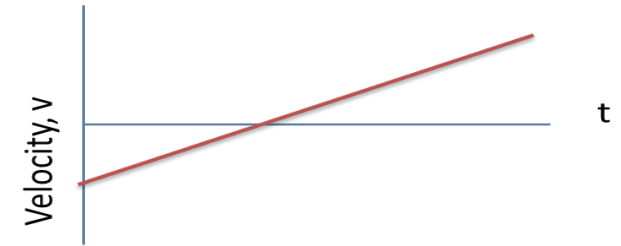
B)

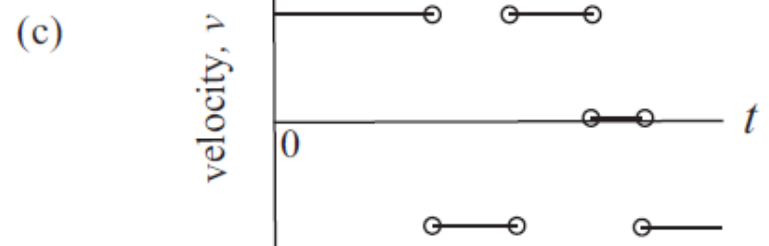
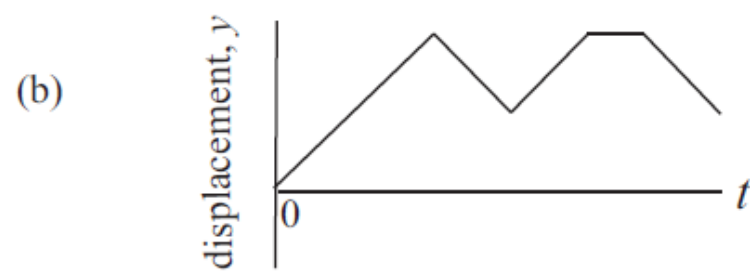
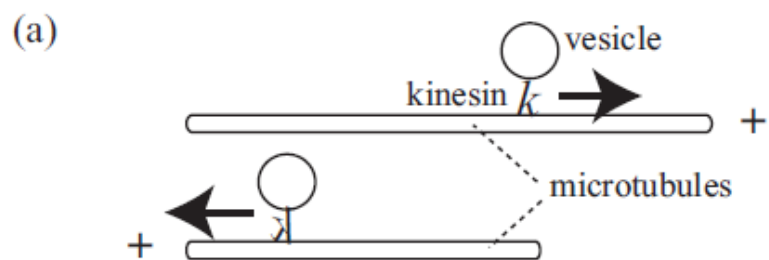
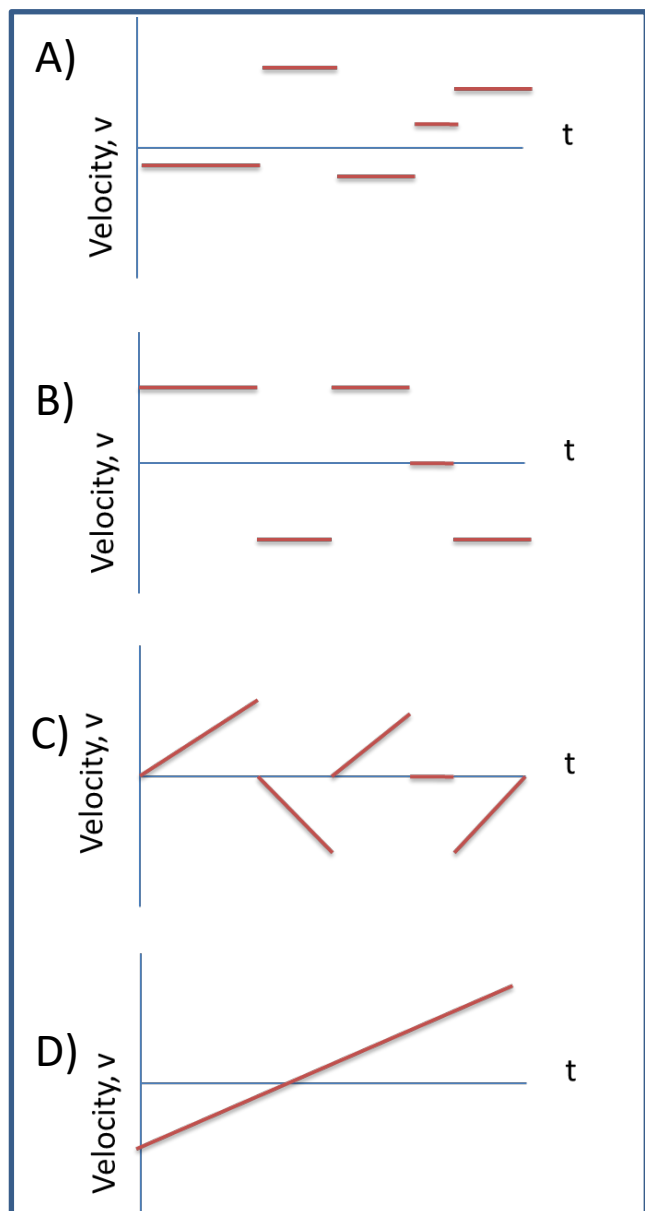


C)



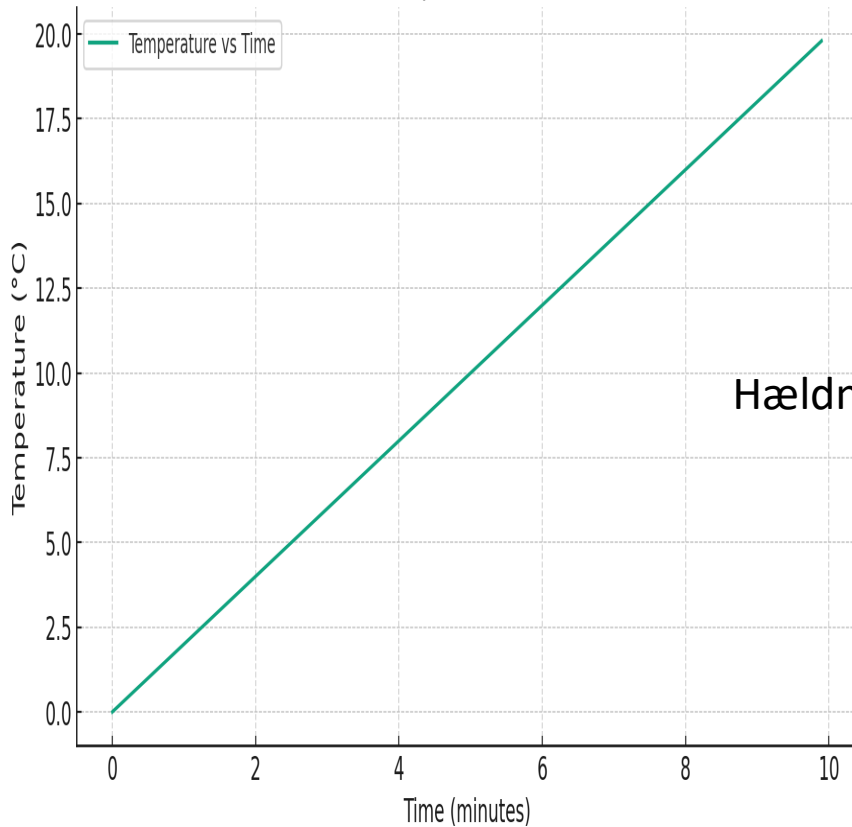
D)





Opvarmning og køling

Linear Increase in Temperature as a Function of Time



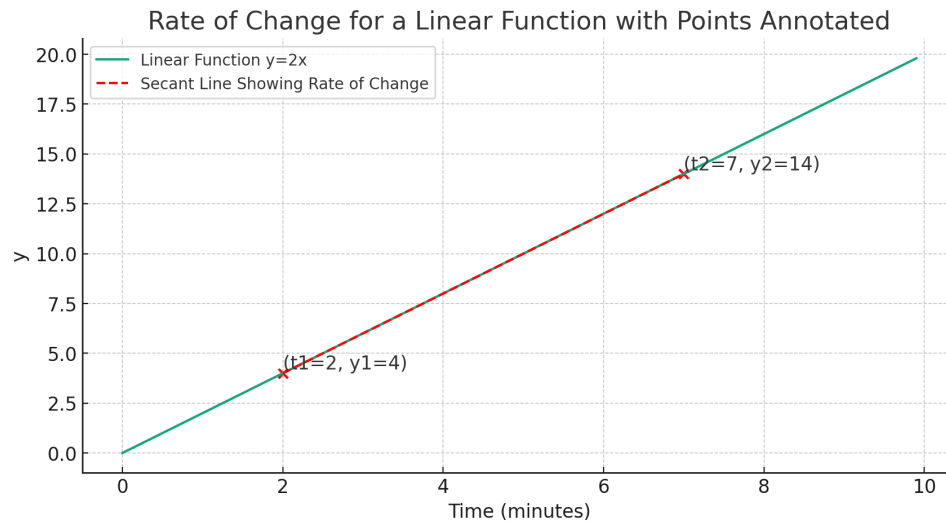
Hvad er ændringshastighed?

$$Y(t) = mt + b$$

Hældningen af en ret linje er en ændringshastighed.

Definition (Ændringshastighed for et lineært forhold).

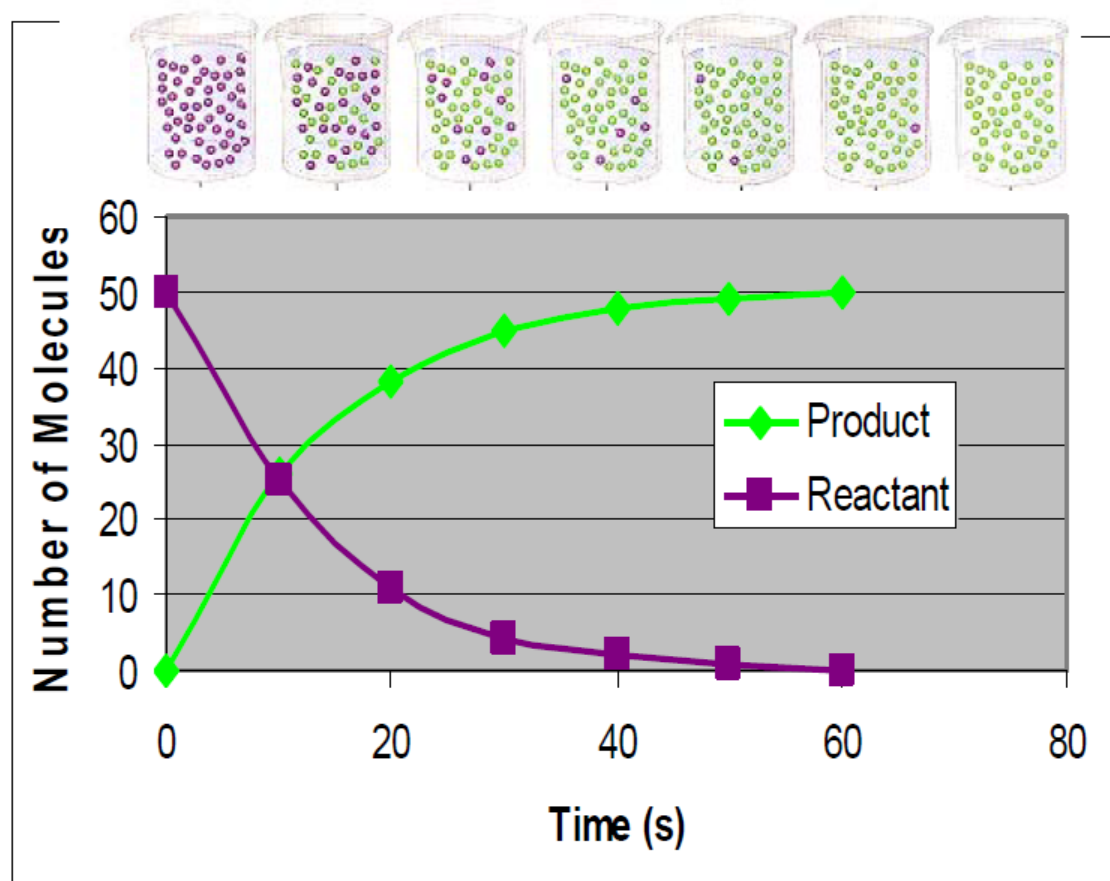
For en ret linje definerer vi ændringshastigheden af y i forhold til tid t som forholdet:



$$\frac{\text{Ændring i } y}{\text{Ændring i } t}$$

$$\frac{\text{Ændring i } y}{\text{Ændring i } t} = \frac{\Delta y}{\Delta t} = \frac{y_2 - y_1}{t_2 - t_1} = \frac{(mt_2 + b) - (mt_1 + b)}{t_2 - t_1} = \frac{mt_2 - mt_1}{t_2 - t_1} = m.$$

Kemiske reaktioner



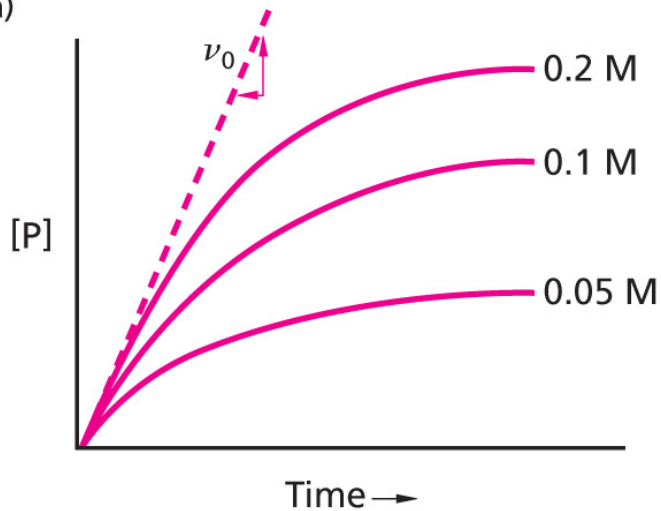
Kemisk kinetik

For den kemiske reaktion $S \xrightarrow{k} P$ er reaktionshastigheden (v) givet ved mængden af produkt per tid.

$$\frac{\Delta[P]}{\Delta t} = v = k[S]$$

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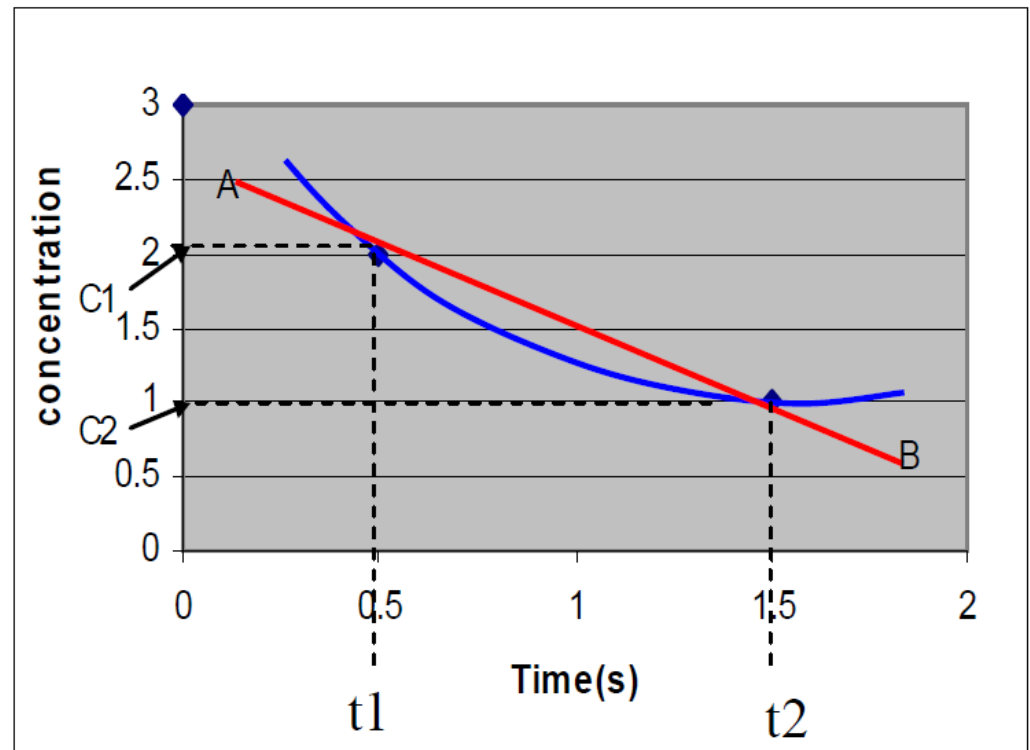
(a)



Hældningen af en sekantlinje er den gennemsnitlige ændringshastighed.

Et plot af koncentration versus tid viser, hvordan Reaktionshastigheden ændres

- Plottet er ikke lineært
 - Raterne ændrer sig over tid
- Den gennemsnitlige rate over tidsintervallet fra t_1 til t_2 er ændringen i [reaktanter] fra c_1 til c_2



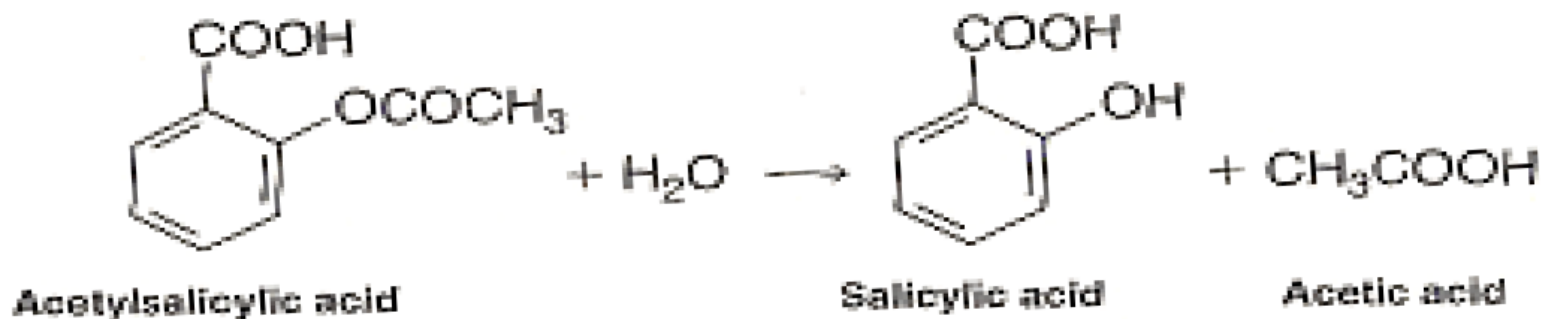
$$\text{Gennemsnitlig rate} = (c_2 - c_1) / (t_2 - t_1) = \text{hældningen af linjen AB} = \Delta c / \Delta t$$

Rate of Reaction

- NOTE: Slope of [reactants] vs. time is always negative
 - Reactants are consumed to form products
 - Notice that the slope of AB is negative
- Rate must always be expressed as positive numbers
- To ensure this, use the absolute value
 - Rate calculated from [reactants] = $\left| \frac{-\Delta C}{\Delta t} \right|$

Lad os se på en eksempelreaktion, som alle har oplevet - brugen af aspirin

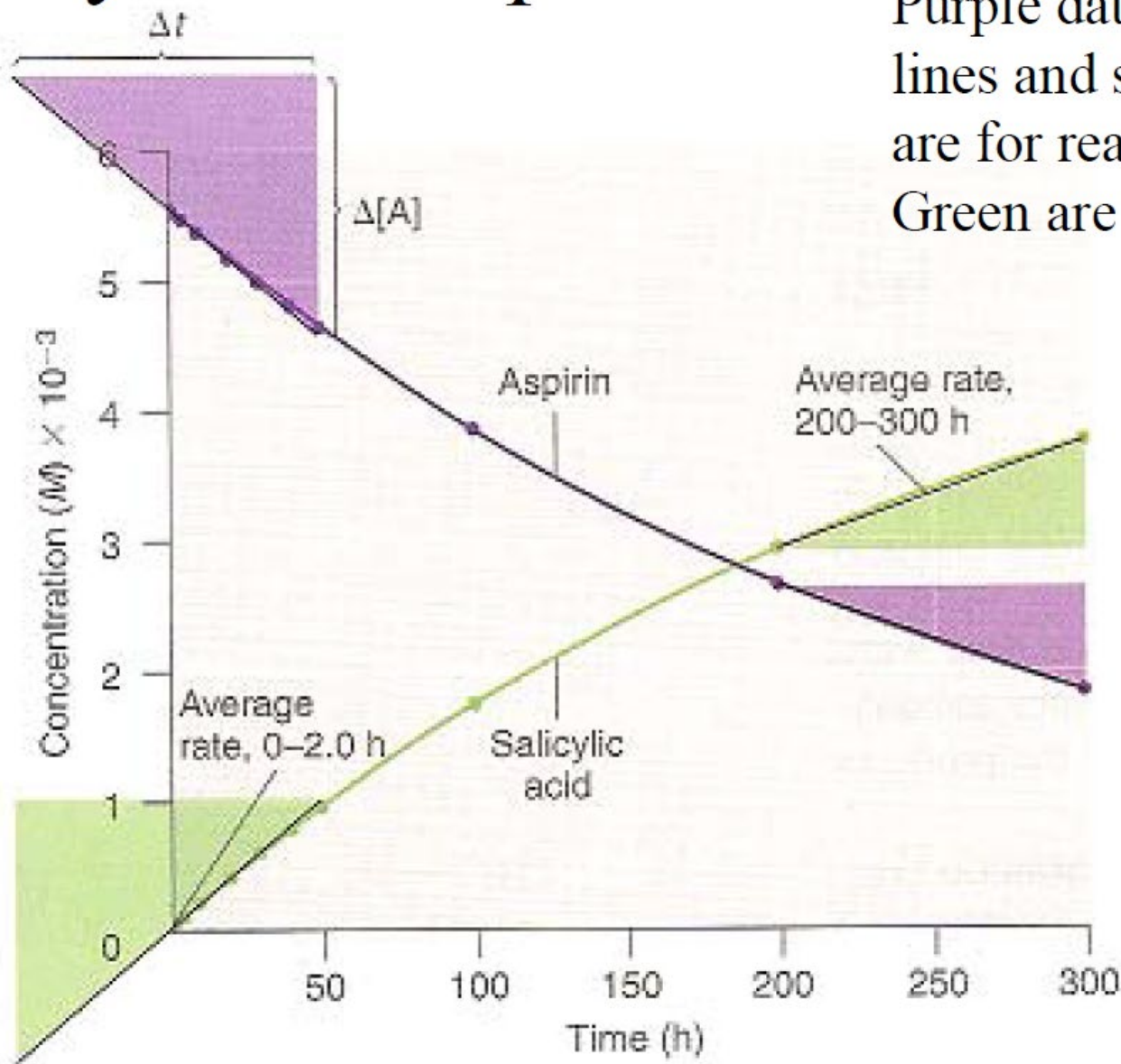
- Aspirin (acetylsalicylsyre) reagerer med vand for at producere salicylsyre og eddikesyre



- Reaktionen sker i din mave og er en hydrolyse reaktion
- Salicylsyre er den egentlige smertestillende og febernedsættende
- Reaktionen blev stoppet på forskellige punkter, så koncentrationen af reaktanter og produkter kunne observeres

Hydrolysis of Aspirin

Purple data markers, lines and shading are for reactants
Green are for products



Tabel 1. Data for hydrolysen af Aspirin i vandig opløsning ved pH 7 og 37C. Koncentrationer i M Find hastigheden for Salicylsyre mellem 200h- 300h.

Time (h)	[aspirin]	[Salicylic acid]
0	5.55×10^{-3}	0×10^{-3}
20	5.15×10^{-3}	0.40×10^{-3}
50	4.61×10^{-3}	0.94×10^{-3}
100	3.83×10^{-3}	1.72×10^{-3}
200	2.64×10^{-3}	2.91×10^{-3}
300	1.82×10^{-3}	3.73×10^{-3}

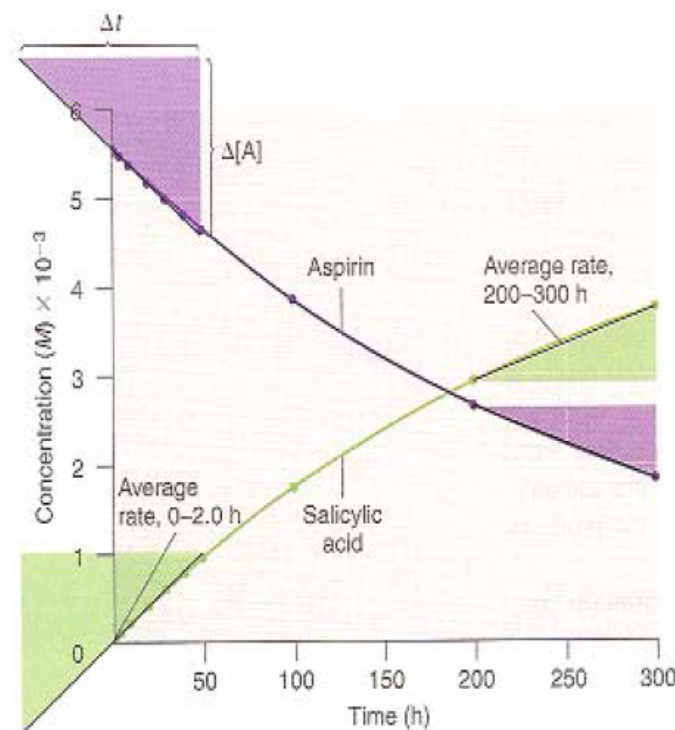
Hydrolysis of Aspirin (near the end of the reaction)

- Towards the end of the reaction, we can figure out how the rate has changed
- For example:

The rate found by the salicylic acid from $t=200$ to $t=300$

$$rate_{(t=200h-300h)} = -\frac{[[salicylic\ acid]_2 - [salicylic\ acid]_0]}{300h - 200h}$$

$$\frac{3.73 \times 10^{-3} M - 2.91 \times 10^{-3} M}{100h} = 8.2 \times 10^{-6} M/h$$



Now calculate the rates for 0-20h for aspirin.
Concentrations are in M

Time (h)	[aspirin]	[Salicylic acid]
0	5.55×10^{-3}	0×10^{-3}
20	5.15×10^{-3}	0.40×10^{-3}
50	4.61×10^{-3}	0.94×10^{-3}
100	3.83×10^{-3}	1.72×10^{-3}
200	2.64×10^{-3}	2.91×10^{-3}
300	1.82×10^{-3}	3.73×10^{-3}

Now how to calculate all the rates for aspirin.

$$\text{Rate calculated from [reactants]} = \left| \frac{-\Delta C}{\Delta t} \right|$$

$$\text{Rate1} = (-[\text{aspirin2}-\text{aspirin1}])/[\text{tid2}-\text{tid1}]$$

$$\text{Rate1} = ([\text{aspirin1}-\text{aspirin2}])/[\text{tid2}-\text{tid1}]=dk1/dt1$$

- $dk1 = [\text{aspirin1}-\text{aspirin2}]$
- $dk2 = [\text{aspirin2}-\text{aspirin3}]$
- $dk3 = [\text{aspirin3}-\text{aspirin4}]$
- $dk = \text{aspirin}[1:5] - \text{aspirin}[2:6]$

$$dt1 = [\text{tid2}-\text{tid1}]$$

$$dt2 = [\text{tid3}-\text{tid2}]$$

$$dt3 = [\text{tid3}-\text{tid3}]$$

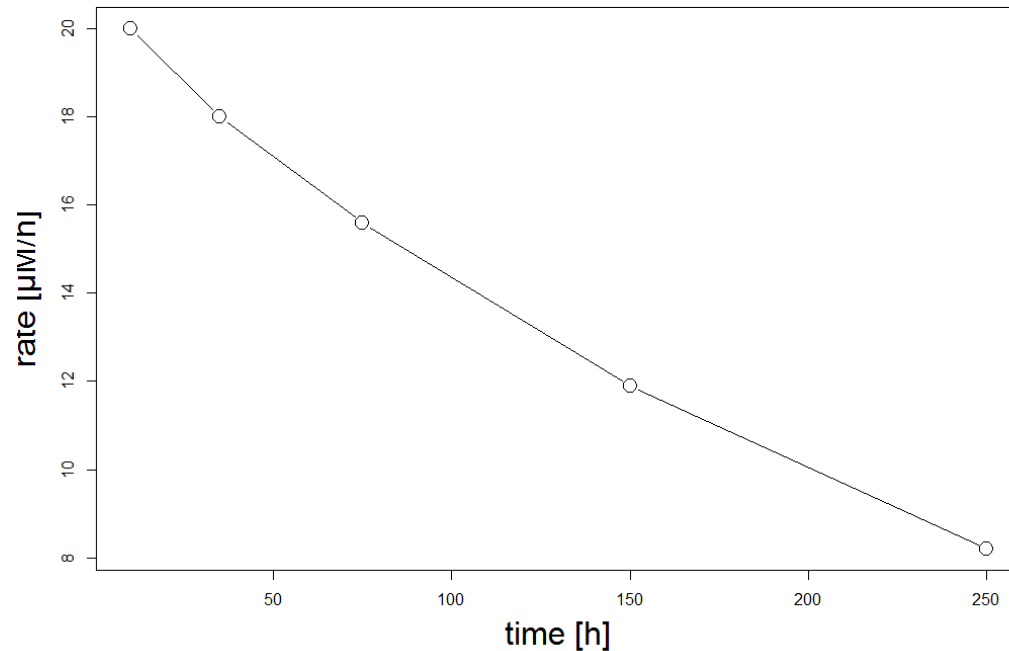
$$dt = \text{tid}[2:6] - \text{tid}[1:5]$$

	Time (h)	[aspirin]
1	0	5.55×10^{-3}
2	20	5.15×10^{-3}
3	50	4.61×10^{-3}
4	100	3.83×10^{-3}
5	200	2.64×10^{-3}
6	300	1.82×10^{-3}

5.15×10^{-3}
4.61×10^{-3}
3.83×10^{-3}
2.64×10^{-3}
1.82×10^{-3}

-

Rate constants



R script

```
t <- c(0,20,50,100,200,300)
k <- c(5.55, 5.15,4.61,3.83,2.64, 1.82)
l=length(t)
```

```
dt=t[(2):l] - t[1:(l-1)]
```

```
dk=k[(2):l] - k[1:(l-1)]
```

```
rate=abs((dk/dt)*1000)
```

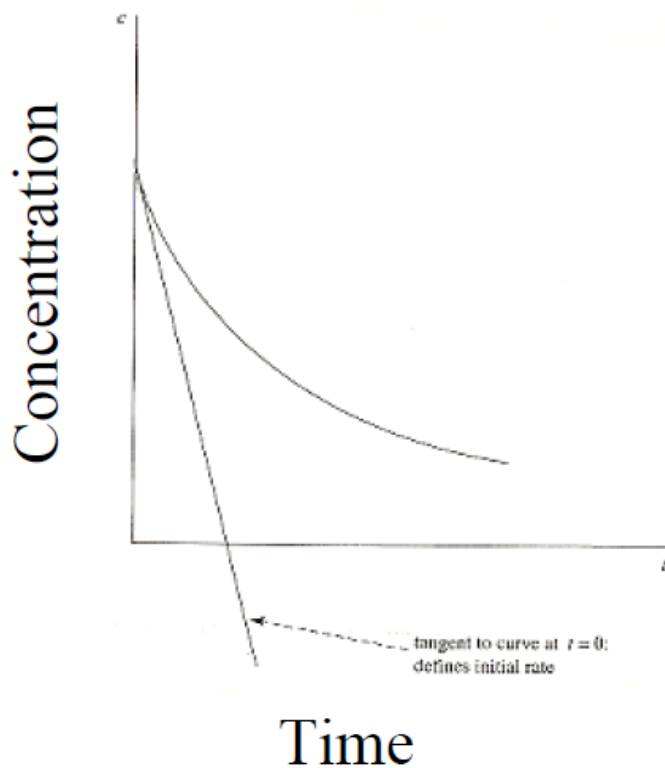
(gange med 1000 for at lave enheder om til
μM/h)

```
tid=(t[(2):l] + t[1:(l-1)])/2
```

```
plot(tid,rate,type="b", xlab = " Tid[h] ",  
ylab = "Rate [μM/h]")
```

The initial rate of the reaction

- Is very important, especially in complex reactions
- Is at the start of the reaction
- Is the line of steepest slope
 - Fastest rate



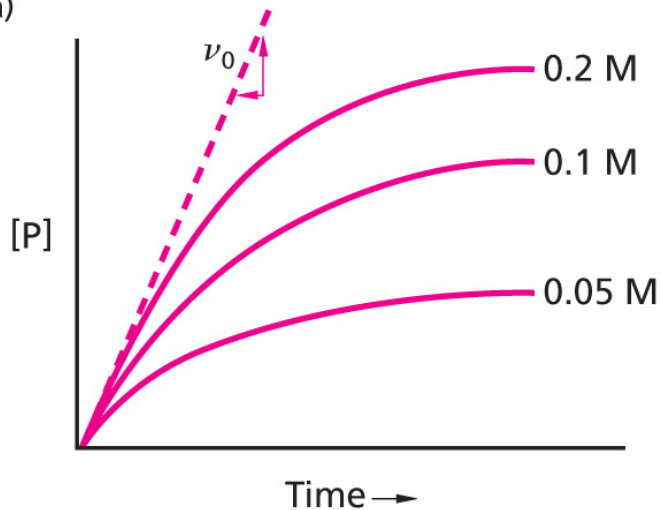
Kemisk kinetik

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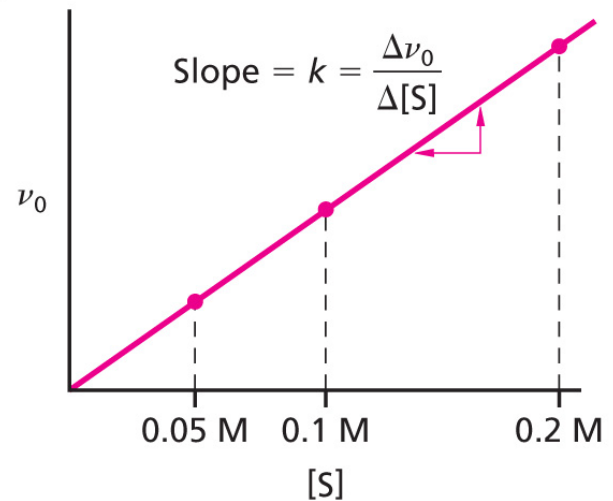
$$\frac{\Delta[P]}{\Delta t} = v = k[S]$$

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(a)

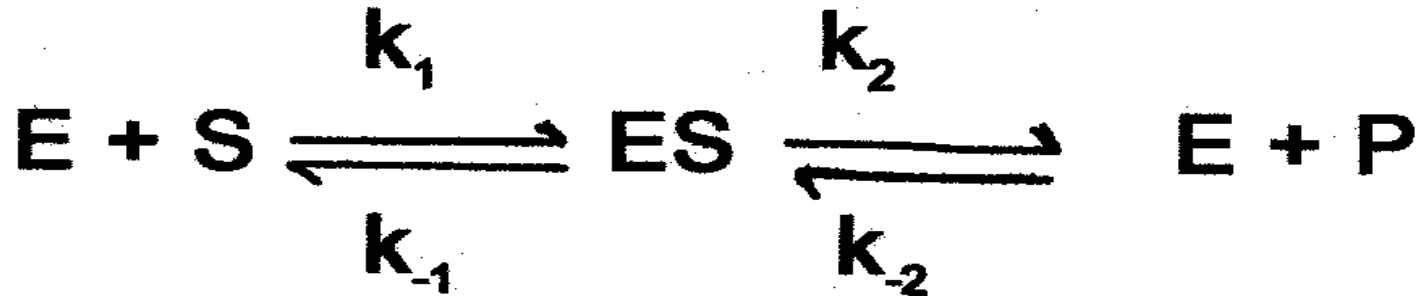
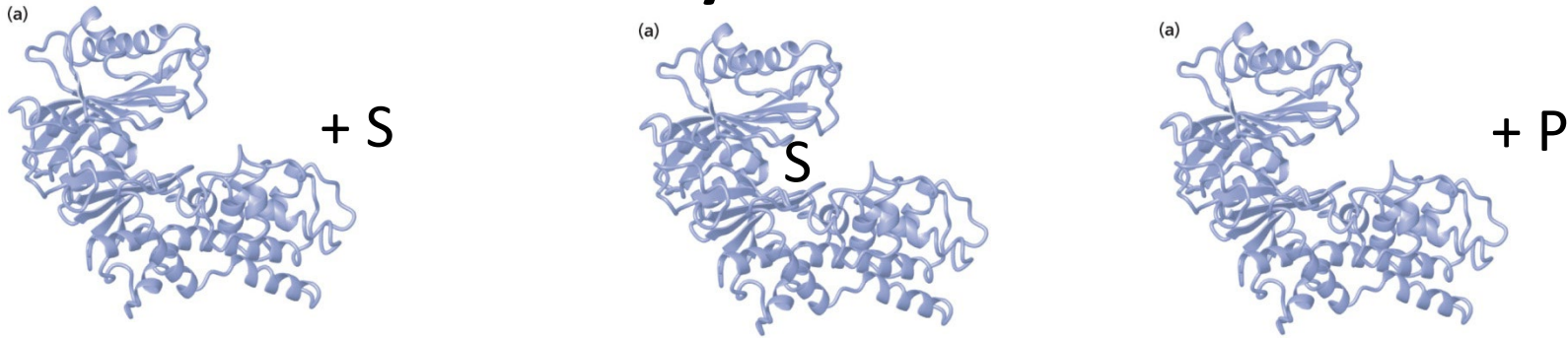


(b)



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Enzyme kinetics



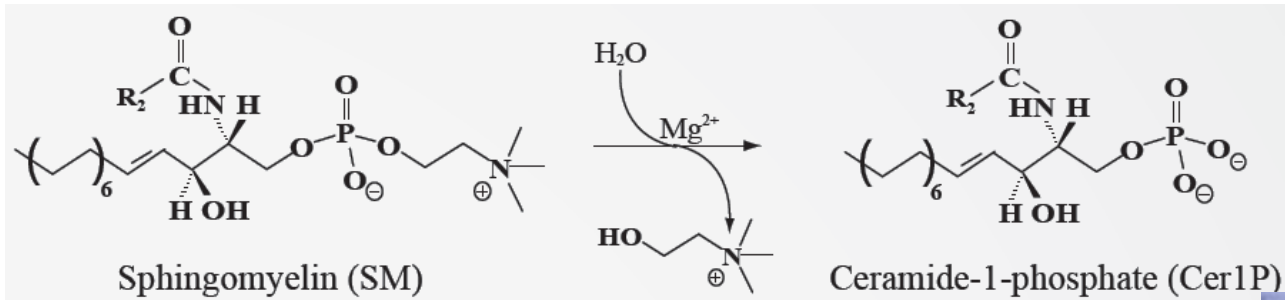
S = substrate P = product

E = enzyme

ES = enzyme-substrate complex

k_1 , k_{-1} , k_2 , k_{-2} are rate constants

Sphingomyelinase D



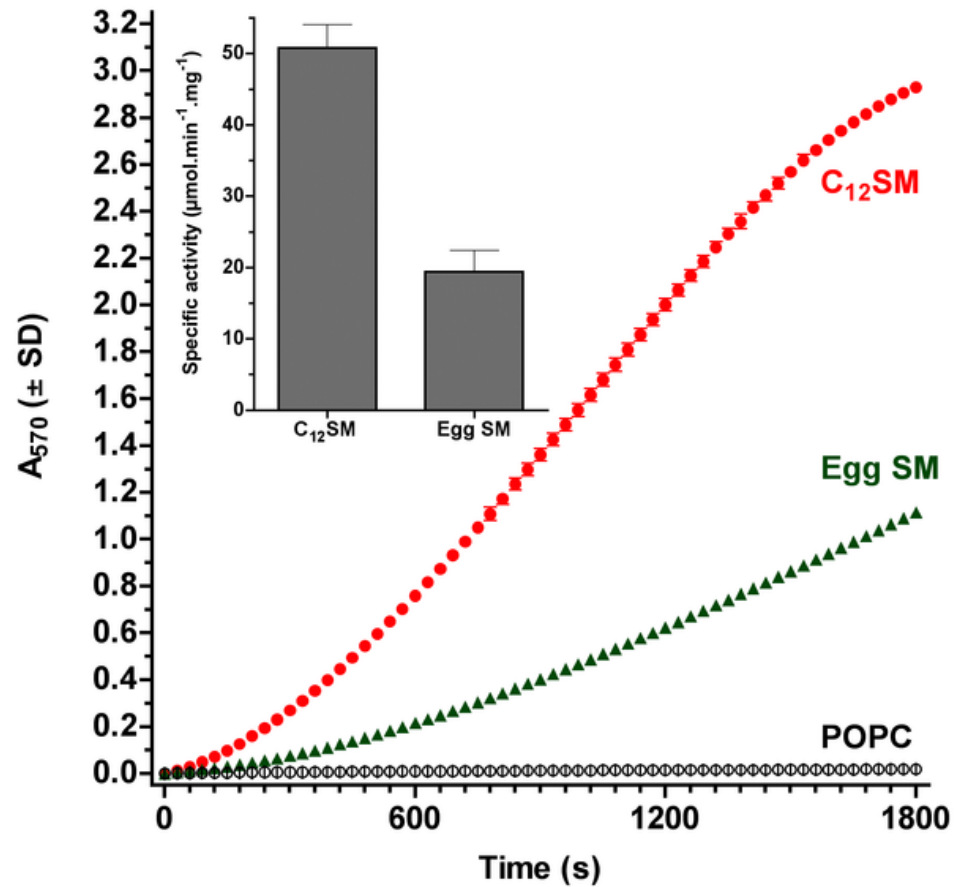
Dermonecrosis



Loxosceles spider

H. Thorke, SDU
J. Brewer, SDU
K. Wagner, SDU
R. Stock, Univ. Autonoma de Mexico

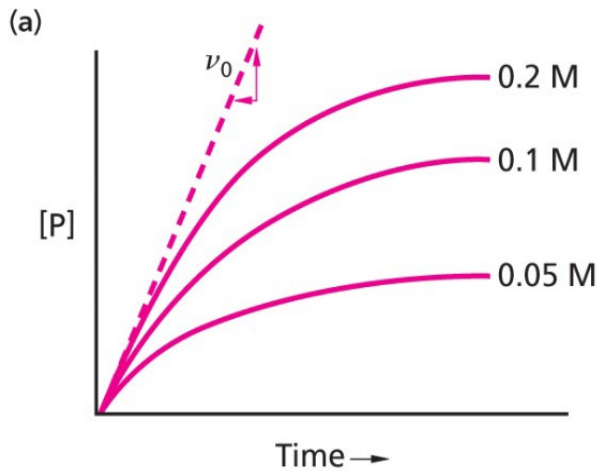
Kinetics of SMD activity on large unilamellar vesicles.



Stock RP, Brewer J, Wagner K, Ramos-Cerrillo B, et al. (2012) Sphingomyelinase D Activity in Model Membranes: Structural Effects of in situ Generation of Ceramide-1-Phosphate. PLoS ONE 7(4): e36003. doi:10.1371/journal.pone.0036003

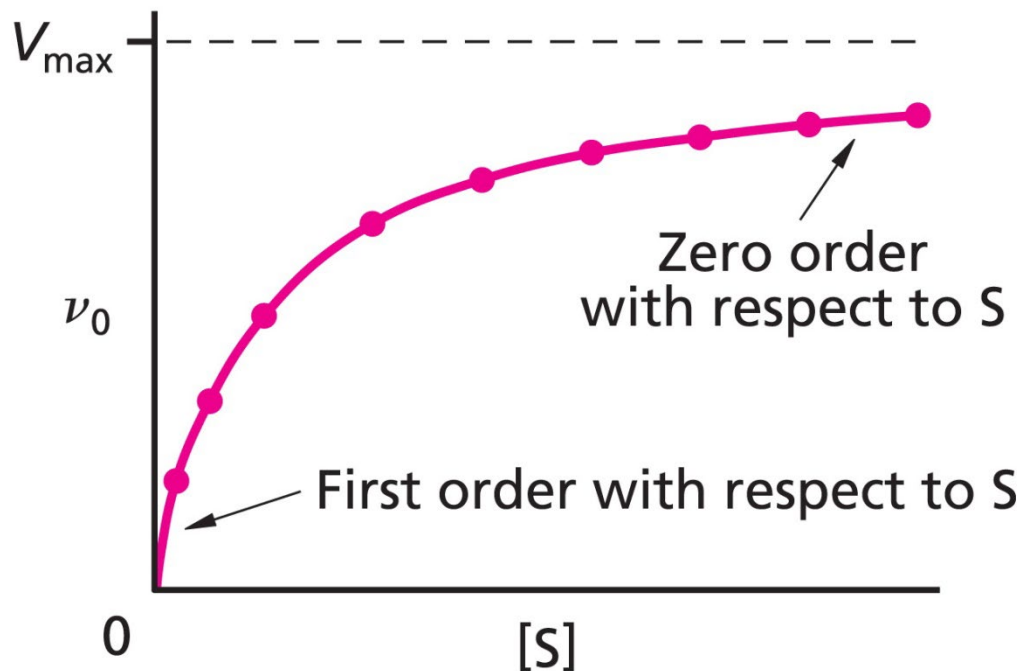
<http://www.plosone.org/article/info:doi/10.1371/journal.pone.0036003>

Initialhastighed for et enzym



v_0 = Initialhastighed

$V_{\max}$: hastighed ved uendelig koncentration S



Første ordens: lineær med S
Nulte ordens: uafhængig af S

Data kan fittes af

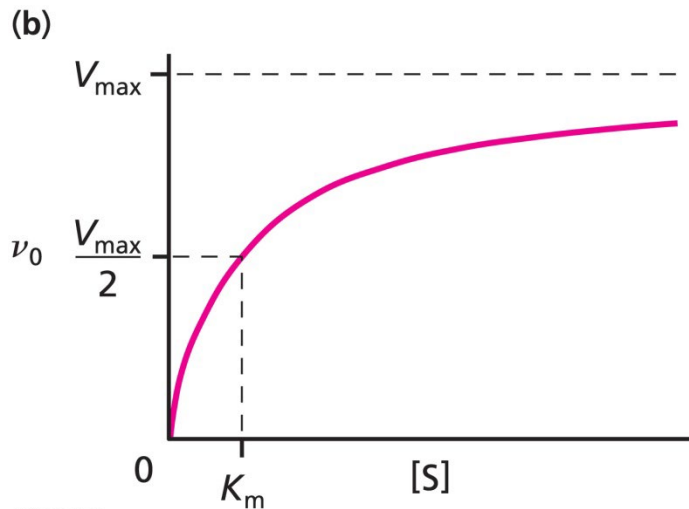
$$y = \frac{ax}{b + x}$$

Ligningen for en rektangulær hyperbel.

Michaelis-Menten Equation

$$v_0 = \frac{V_{\max}[S]}{K_m + [S]}$$

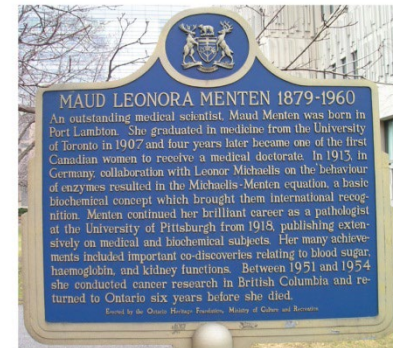
K_m er Michaelis konstanten . K_m koncentrationen af S når $v_0 = V_{\max}/2$



L. Michaelis 1875-1948



M. Menten 1879-1960



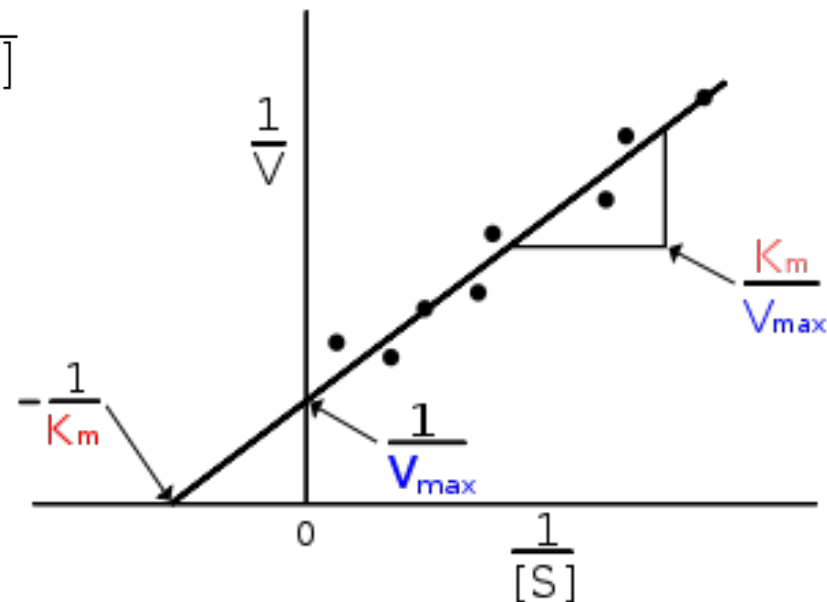
Determination of Kinetic Parameters

$$v_0 = \frac{V_{max}[S]}{([S] + K_m)}$$

$$\frac{1}{v_0} = \frac{([S] + K_m)}{V_{max}[S]} = \frac{[S]}{V_{max}[S]} + \frac{K_m}{V_{max}[S]}$$

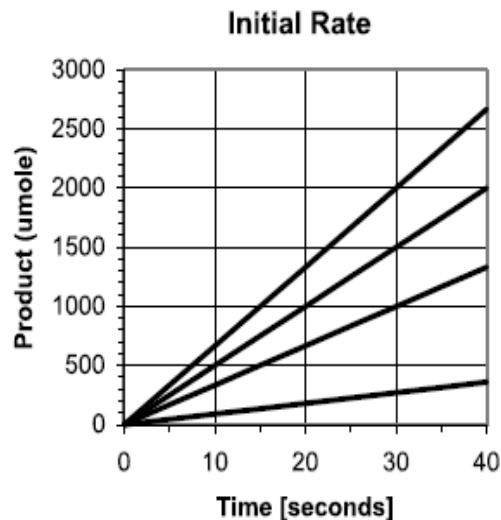
$$\frac{1}{v_0} = \frac{K_m}{V_{max}} \frac{1}{[S]} + \frac{1}{V_{max}}$$

Linearized double-reciprocal plot is good for analysis of two-substrate data or inhibition



Example

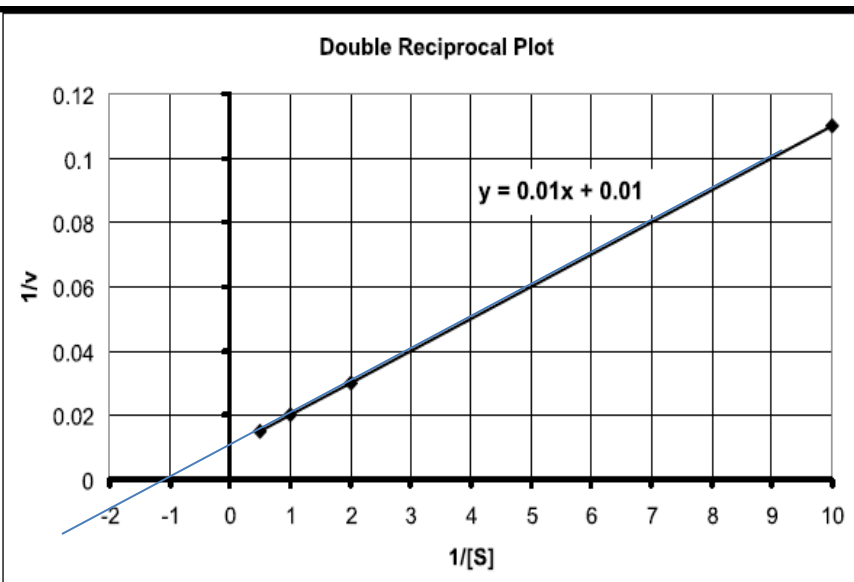
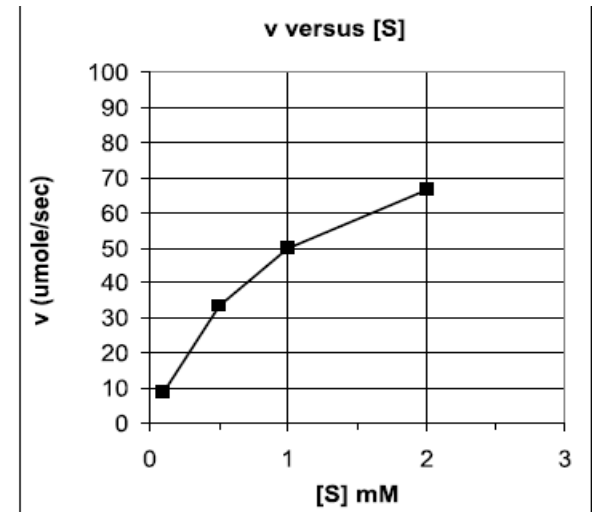
- The following graph shows the amount of product produced as a function of time for four different concentrations of substrate, 0.1 mM, 0.5 mM, 1 mM, and 2 mM. The amount of enzyme present in the reaction was 1 nmol.
 - a) Determine K_M and V_{max} from these data.



[S]	Product Formed/30 sec
0.1 mM	270 umole
0.5 mM	1000 umole
1 mM	1500 umole
2 mM	2000 umole

Example continued

[S]	Product Formed/30 sec	$v_i = \Delta P / \Delta t$
0.1 mM	270 umole	9.1 umole/sec
0.5 mM	1000 umole	33 umole/sec
1 mM	1500 umole	50 umole/sec
2 mM	2000 umole	66 umole/sec



V_{max} : The y-intercept of this plot is 0.01 sec/umole.
Therefore V_{max} is 100 umole/sec.

K_M : The x-intercept of the plot is -1.
Therefore $K_M = 1$ mM.

Rater: Gennemsnitlige vs. øjeblikkelige

- Gennemsnitlige rater er over en periode
 - Giver begrænset information
- Når tidsintervallerne bliver mindre og mindre, nærmer de sig et bestemt "øjeblik"
- Koncentrationen på det tidspunkt i forhold til tiden kaldes den øjeblikkelige rate